

Project Proposal: Telco Customer Churn Prediction

Dataset: [IBM Telco Customer Churn](#)

1. Introduction

The telecommunications industry operates in a highly competitive environment in which customers can choose among multiple service providers offering similar internet, telephone, and digital services. One recurring challenge for telecommunications companies is **customer churn**, which refers to customers ending or discontinuing their relationship with a service provider. Identifying customers who are likely to leave is important because customer loss can affect recurring revenue and requires companies to allocate additional resources toward attracting replacement subscribers.

Telecommunications companies collect different types of customer information, including demographic characteristics, subscribed services, contract arrangements, payment methods, tenure, and billing information. These data can be analyzed through data science and machine learning to identify patterns associated with customer churn and estimate the probability that a customer may discontinue service.

This study proposes the development of a machine learning-based customer churn prediction system using the **Telco Customer Churn dataset**, which is publicly available through Kaggle and originates from IBM sample data. The dataset contains **7,043 customer records and 21 columns**, with each row representing one customer. Its variables describe customer demographics, subscribed telecommunications services, account information, billing characteristics, and whether the customer churned during the previous month.

The project will follow the eight stages of the Data Science Lifecycle: **Problem Definition, Data Collection, Data Cleaning and Preprocessing, Exploratory Data Analysis, Modeling, Evaluation, Deployment, and Communication**. Several machine learning algorithms will be developed and compared to determine their effectiveness in predicting churn. The selected model will then be incorporated into an interactive Streamlit application that can display customer churn probabilities, model explanations, and analytical visualizations.

Rather than treating machine learning simply as a method for producing a prediction, the project will also investigate the characteristics associated with customer churn. This can help transform model predictions into more understandable information that may support customer-retention decision-making.

2. Statement of the Problem

- **Problem Definition:**

Telecommunications companies manage large numbers of subscribers with different contract types, service subscriptions, billing arrangements, and customer histories. Although customer information is available, it may be difficult to manually determine which combination of characteristics is associated with a greater likelihood of customer churn.

The central problem addressed by this study is therefore:

How can customer demographic, service, contractual, and billing information be used to predict whether a telecommunications customer is likely to churn?

From a data science perspective, the problem will be treated as a **supervised binary classification problem** in which:

- **Churn = Yes** represents a customer who discontinued service.
- **Churn = No** represents a customer who remained with the company.

The model will use the available customer characteristics as predictor variables and **Churn** as the target variable.

- **The Gap**

Customer churn prediction has already been widely investigated using machine learning techniques. Therefore, the purpose of this project is not to claim that churn prediction itself is an unexplored problem. Instead, the project addresses the practical challenge of determining which classification method provides an appropriate combination of predictive performance and interpretability for the selected telecommunications dataset.

Another important challenge is the distribution of the target variable. The IBM Telco dataset contains approximately **26.5% churn customers and 73.5% non-churn customers**, meaning that the classes are not equally represented. A model that predicts the majority class too frequently could obtain relatively high overall accuracy while still failing to identify many customers who actually churn.

For this reason, the project will examine several performance measures, including precision, recall, F1-score, ROC-AUC, and the confusion matrix, rather than evaluating models solely through accuracy. Recent telecommunications churn research similarly emphasizes the importance of evaluating several metrics when working with imbalanced classification problems.

The project will additionally examine how model predictions can be interpreted. A prediction that a customer has a high probability of churn may be more useful when the system can also identify the factors that contributed to that prediction.

3. Significance of the Study

- **Importance:** Early identification of at risk customers allows companies to deploy targeted retention incentives for example discounts, and contract upgrades on where they are needed.

- **Beneficiaries:**

- **Telecom Management & Retention Teams:** To optimize marketing and retention budgets.

Managers and retention teams may use churn-risk information to better understand which customer segments are associated with higher attrition. These insights could support the allocation of retention resources and the design of targeted customer-retention strategies.

- **Customer Service Representatives:** To prioritize outreach to high risk accounts.

Customer service teams could use a churn-risk system to identify customers who may require additional attention. The model's explanations may also indicate which characteristics contributed to the customer's predicted risk.

- **End-User Consumers:** Who may receive better targeted offers and improved customer satisfaction.

- **Data Analysts and Business Decision-Makers**

The study can demonstrate how demographic, contractual, service, and financial variables relate to customer churn. Analysts may use these findings to support future customer-segmentation studies and evidence-based business decisions.

- **Students and Researchers**

The project provides a practical demonstration of an end-to-end classification problem involving preprocessing, exploratory analysis, class imbalance, model comparison, explainable machine learning, and deployment.

4. Research Questions

1. How can customer attrition be structured as a supervised binary classification problem, and what preprocessing strategies are needed to address the class imbalance between churned and non-churned customers in the IBM Telco dataset?
2. What data quality issues, missing values, or formatting inconsistencies (such as in **TotalCharges**) exist in the raw dataset, and how can they be resolved to ensure feature integrity?
3. What primary trends, statistical distributions, and bivariate correlations emerge between customer attributes (such as tenure, monthly charges, and contract type) and churn behavior?

4. Which machine learning classification algorithm (e.g., Logistic Regression, Random Forest, Gradient Boosting) achieves the optimal balance of predictive performance when evaluated across metrics like precision, recall, F1-score, and ROC-AUC?
5. How effectively can an interactive web application integrate the trained model and feature interpretations to deliver real-time churn risk predictions and actionable insights for stakeholders?

5. Objectives

1. Formulate the business problem as a binary classification task and implement data preprocessing techniques (such as class weighting or resampling) during the early lifecycle stages to handle class distribution skewness.
2. Audit, clean, and transform the raw customer data from Kaggle to correct anomalies, handle missing values, and prepare clean categorical and numerical features for analysis.
3. Conduct thorough **Exploratory Data Analysis (EDA)** using descriptive statistics and visualizations to uncover key demographic and contractual risk factors associated with customer attrition.
4. Train, tune, and rigorously evaluate multiple classification models using multi-metric performance criteria to identify the most effective predictive algorithm.
5. Build and launch an interactive **Streamlit web application** and document the findings in a comprehensive report to support customer-retention decision-making.

6. Methodology

- a. This study will use the IBM Telco Customer Churn dataset, obtained from Kaggle. The dataset contains 7,043 customer records and 21 columns, with information about customer demographics, services, contracts, tenure, payment methods, and billing. The target variable is Churn, which indicates whether a customer discontinued the service.

The dataset is appropriate for this project because it contains both numerical and categorical customer attributes that can be analyzed to identify patterns related to churn. Before modelling, the data will be checked for missing values, duplicate records, inconsistent data types, and other data quality issues. Categorical variables will be encoded into numerical values, while numerical variables will be prepared for modeling as needed

- b.
- c. The selected model will be presented in a prototype Streamlit application. Users will be able to enter customer information and view the model's predicted churn probability, along with relevant visualizations and an explanation of influential features. The app will demonstrate how predictions and findings could be communicated to the stakeholders.

7. Application of the Data Science Lifecycle

